MILITARY SPECIFICATION

# CARTRIDGE, CALIBER .38, SPECIAL, BALL, M41A1

## 1. SCOPE

1.1 This specification covers a cartridge intended for firing in Revolvers Caliber .38 Special, 2 inch or 4 inch barrel procured in accordance with D-R-1187.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-636 | - Visual Inspection Standards for Small Arms Ammunition through Caliber .50 |
| MIL-STD-644 | - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition |
| MIL-STD-1168 | - Lot Numbering of Ammunition |

DRAWINGS

U.S. Army Munitions Command

| | |
|---|---|
| C11735596 | - Cartridge, Caliber .38, Special, Ball, M41A1 |
| F7553763 | - Packing and Marking; Cartridges, Caliber .38, Special, Ball, Cartons; Box, Ammunition, M2A1; Box, Wirebound |

FSC 1305

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2.1 (Cont'd)

DRAWINGS

| | |
|---|---|
| C7643674 | - Classification of Cartridge Case Defects |
| LI-11744170 | - Inspection Equipment List for Cartridge, Caliber .38, Special, Ball, M41 |

PUBLICATIONS

U.S. Army Munitions Command
TECP 700-700 Vol. III - Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with the requirements specified on Drawing C11735596, referenced specifications and the following:

3.2 Manufacturing Process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.2)

3.3 Bullet extraction.-The force required to extract the bullet from the cartridge case shall be not less than 130 pounds.

3.4 Residual stress.-The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.5 Velocity.-The average velocity of the sample cartridges, conditioned at 68° to 72° Fahrenheit (F) shall be 950 feet per second (ft/sec) plus or minus 25ft/sec at 15 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 27 ft/sec.

3.6 Chamber pressure.-The chamber pressure of the cartridges, temperature conditioned and fired at the temperature prescribed, shall not exceed the following:

| Temperature | Average Pressure | Single Shot Pressure |
|---|---|---|
| Degree (°) Fahrenheit (F) | Pounds per square inch (PSI) Maximum | (PSI) Maximum |
| -40° to -45° | 20,000 | 24,000 |
| 68° to 72° | 16,000 | 20,000 |
| 125° to 130° | 20,000 | 24,000 |



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3.7 Accuracy.-The average of the diagonals (D) of all targets of the sample cartridges fired at 50 yards shall not exceed 7.46 inches.

3.8 Function and casualty.-The cartridge shall function without casualty at ambient temperatures, at minus 40 degrees to minus 45 degrees F and at plus 125 degrees to plus 130 degrees F.

3.9 Waterproof.-The cartridge shall not release more than one air bubble when subjected to a pressure differential of 5 psi for 15 seconds.

3.10 Cartridge elongation.-The cartridge chambered but not fired, shall not, after the firing of other cartridges chambered in the same cylinder, exceed 1.262 inches in length.

3.11 Primer sensitivity.-The energy imparted by a steel ball 1.94 ounces falling 10 inches shall cause initiation of the primer. The energy imparted by a steel ball 1.94 ounces falling 2 inches shall not cause initiation of the primer.

3.12 Propellant charge.-Each cartridge shall contain a propellant charge weight of 3 grains minimum when loaded with HPC 18 type propellant.

3.13 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.13.1 Metal defects.-The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.13.2 Foreign matter.-The cartridges shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.



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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 3400 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only, and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 The size of an ammunition lot shall be not less than 200,000 nor more than 500,000 cartridges.

4.3.1.3 Lot identification.-Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number in accordance with MIL-STD-1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.-Each charged cartridge case shall be examined for propellant charge, (wt) minimum, requirement of 3.12. Method of examination shall be approved by the contracting officer. The equipment design shall provide control(s) that will assure rejection of defective propellant charge (wt), in the cartridge case. One hundred per cent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 per cent and the AQL for the minor class shall be 1.50 per cent.

4.3.2.1 Classification of defects.-The classification of defects shall be as specified in Table I.



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Table I

| No. | Defect and Method of Inspection | Critical | Major | Minor | Major or Minor |
|---|---|---|---|---|---|
| | Visual 1/ | | | | |
| 1 | Cartridge | | | | |
| | Discolored, dirty, oily, smeared | | | x | |
| 2 | Corroded; or stained if etched | | x | | |
| | Case | | | | |
| 4 | Round head | | | x | |
| 5 | Dent | | | | x |
| 6 | Split case | | | | |
| | in K, L, or M location | x | | | |
| | in I, S, or J location | | x | | |
| 7 | Perforated case | x | | | |
| 8 | Draw scratch | | | | x |
| 9 | Scratch(case) | | | x | |
| 10 | Case not crimped in cannelure | | x | | |
| 11 | Scaly metal (case) | x | | | |
| 12 | No chamfer on head (rim) | | | x | |
| 13 | Wrinkle | | | x | |
| 14 | Bulge | | | x | |
| 15 | Illegible or missing head stamp | | | x | |
| 16 | Defective head | | | x | |
| 17 | Defective mouth | | | x | |
| | Bullet | | | | |
| 18 | Dent | | | x | |
| 19 | Scratch | | | x | |
| 20 | Split bullet jacket | | x | | |
| 20A | Defective cannelure | | x | | |
| 21 | Loose bullet | | x | | |
| 22 | Scaly metal | | | | x |
| 23 | Upset (crooked) point | | | x | |
| 24 | Exposed steel (clad jacket) | | | x | |
| 25 | Flat point | | | x | |



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Table I (Cont'd)

| No. | Defect and Method of Inspection | Critical | Major | Minor | Major or Minor |
|---|---|---|---|---|---|
| | Primer | | | | |
| 26 | No primer | x | | | |
| 27 | Cocked Primer | x | | | |
| 28 | Inverted primer | x | | | |
| 29 | Loose primer | | x | | |
| 30 | Nicked or dented primer | | | x | |
| 31 | No waterproofing (primer pocket joint) | | | x | |
| | Gaging | | | | |
| | Cartridge | | | | |
| 32 | Total length, max. | | x | | |
| 33 | Total length, min. | | | x | |
| 34 | Cartridge profile failure (requiring more than 5 pounds dead weight to insert in profile and alignment gage) | | x | | |
| | Case | | | | |
| 35 | Diameter of head | | | x | |
| 36 | Thickness of head, max. | | x | | |
| 37 | Thickness of head, min. | | | x | |
| | Primer | | | | |
| 38 | Depth of primer, min. | | x | | |
| 39 | Depth of primer, max. | | | x | |

1/ Refer to Caliber .45 section of Standard MIL-STD-636 for Visual Standards for defects 1 through 31. In the event of conflict between Table 1 of this specification and Standard MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.3.2.2 Packaging, packing and marking inspection.-During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy or blank cartridge shall be classed as a critical defect. Occurrence of any type other than those listed shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.



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4.3.3 Tests.-The tests listed in Table II and 4.3.3.1 shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Prior to performance of the tests cited in Table II a sample of cartridges shall be tested for primer sensitivity. The sample shall be selected from the entire cartridge lot or from individual sub-lots, at the option of the contractor, provided that the size of each sub-lot is not less than 200,000 cartridges. The test sample shall be representative of the entire cartridge lot or sub-lot as applicable, and shall consist of the number of cartridges required for performance of a complete run-down test wherein fifty primers are tested at each height. If the average critical height ($\bar{H}$) plus five standard deivations (5 sigma) exceeds 10 inches, or if the average critical height ($\bar{H}$) minus two standard deviations (2 sigma) is less than 1 inch, the lot or sub-lot of cartridges shall be rejected subject to performance of a second complete run-down test wherein one hundred primed cases are tested at each height. Failure of the second sample to comply with the above criteria shall be cause for rejection of the lot or sub-lot of cartridges as applicable.

4.3.3.2 Test samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into test samples.

Table II

| Test | Number of Cartridges -40 to -45 | Ambient | 125 to 130 | Requirement Paragraph |
|---|---|---|---|---|
| Bullet extraction 1/ | | 20 | | 3.3 |
| Residual stress (mercurous nitrate) | | 20 | | 3.4 |
| Velocity 2/ | | 20 | | 3.5 |
| Chamber pressure 2/ | 20 | 20 | 20 | 3.6 |
| Accuracy 2/ | | 100 | | 3.7 |
| Function and casualty 3/ | | | | 3.8 |
| Revolver, Caliber .38 Special 4 inch barrel (Colt) | 408 | 408 | 408 | |
| Revolver, Caliber .38 Special 4 inch barrel (S & W) | 408 | 408 | 408 | |
| Waterproof 4/ | | 20 | | 3.9 |
| Cartridge elongation 1/ | | 10 | | 3.10 |
| Propellant charge (wt) 5/ | | 315 | | 3.12 |



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4.3.3.2 (Cont'd)

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the initial test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of the cartridges in any sample to comply with the applicalble requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the initial test for the temperature or temperatures at which the failure occured. Failure of the cartridges of the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ The lot shall be rejected when the function and casualty defects plus all other firing defects exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the initial test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantities specified under function and casualty test shall be fired in all the service weapons specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occured in the initial test. If the total number of defects in the combined first and second samples exceed the acceptance number for the cumulative sample, the lot shall be rejected. If in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance for the cumulative sample, the lot shall be rejected.

4/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four but less than nine cartridges fail in the initial test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample none or more cartridges fail to comply with the applicable requirement.

5/ If one or more cartridges contain less than the minimum required propellant weight, the lot shall be rejected.

4.3.3.3 Firing defect.-Firing defects and acceptance numbers shall be as specified in Table III.



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4.3.3.3 (Cont'd)

Table III

| Defects | Acceptance Number First Sample | Acceptance Number Cumulative First ande Second Sample |
|---|---|---|
| 1. Misfire 6/ | | |
| a. No vent hole, or obstruction in vent area | 0 | - |
| b. Other | 1 | 2 |
| 2. Bullet remaining in bore 1/ | 0 | - |
| 3. Primer leak | | |
| a. Perforation in firing pin indent in primer cup | 19 | 45 |
| b. Escape of gas through primer cup other than a. | 13 | 29 |
| c. Escape of gas around primer cup 2/ | 50 | 165 |
| d. Loose primer 3/ | 19 | 45 |
| e. Blown primer 4/ | 0 | 1 |
| 4. Case casualty | | |
| a. Longitudinal split 5/ | | |
| (1) Mouth (I) | 38 | 95 |
| (2) Body (J) | 28 | 69 |
| (3) Body (K) | 1 | 2 |
| (4) To head (L) | 0 | 1 |
| (5) Through head (M) | 0 | 1 |
| b. Circumferential rupture 5/ | | |
| (1) Partial (J) | 7 | 16 |
| (2) Partial body (K) | 0 | 1 |
| (3) Partial head (L) | 0 | 1 |
| (4) Complete | 0 | 1 |
| 5. Weapon stoppage 7/ | 0 | 1 |

1/ No retest permitted. Lot shall be rejected.

2/ Gas escapes around more than 50 percent of periphery of cup.

3/ Independent movement of primer in case pocket or primer falls out of pocket.

4/ Primer not in pocket after firing and both head of cartridge case and pocket enlarged and distorted (determined visually).

5/ For location of defects indicated by letters in parentheses, see Drawing C7643674.



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6/ Each cartridge that misfires shall be disassembled and examined for presence of a vent hole in the primer pocket, or any obstruction in the vent hole area of the primer pocket that can be assignable as the cause from misfire. If the vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted.

7/ Weapon stoppage, with the exception of misfire, observed in all tests shall be included.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed in LI-11744170, except as provided in 4.3.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.-The test shall be conducted in accordance with Appendix "A". The rate of travel of the test head, in the bullet extractor machine, shall be not less than three (3) inches nor more than six (6) inches per minute.

4.4.2 Residual stress (mercurous nitrate).-The test shall be conducted in accordance with Appendix "B".

4.4.3 Velocity.-The test shall be conducted in accordance with Appendix "C".

4.4.4 Chamber pressure.-The test shall be conducted in accordance with Appendix "D". The obturators shall be of the wax-filled type (see 6.3).

4.4.5 Accuracy.-The test shall be conducted in accordance with Appendix "E".

4.4.6 Function and casualty.-The test shall be conducted in accordance with Appendix "F". The cartridges shall be fired single action and double action in the weapons & temperature as listed below:

Revolver, Caliber .38 Special, 4 Inch Barrel (Colt)

| Temperature F | $-40^{\circ}$ to $-45^{\circ}$ | Ambient | $125^{\circ}$ to $130^{\circ}$ |
|---|---|---|---|
| Single action | 204 | 204 | 204 |
| Double action | 204 | 204 | 204 |

Revolver, Caliber .38 Special, 4 Inch Barrel (S & W)

| Temperature F | $-40^{\circ}$ to $-45^{\circ}$ | Ambient | $125^{\circ}$ to $130^{\circ}$ |
|---|---|---|---|
| Single action | 204 | 204 | 204 |
| Double action | 204 | 204 | 204 |

4.4.7 Temperature (High and Low).-The cartridges for the high and low temperature test shall be conditioned in accordance with Appendix "G".



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4.4.8 Waterproof.-The test shall be conducted in accordance with Appendix "H".

4.4.9 Cartridge elongation.-The test shall be conducted in accordance with Appendix "I". The cartridges used in this test shall be scrapped.

4.4.10 Propellant weight.-The bullet of the test sample shall be removed from the cartridges, then the propellant from each case shall be weighed. Weights shall be taken on a balance scale that is calibrated to the nearest 0.01 of a grain. Results of each propellant weight shall be recorded for Government verification.

4.4.11 The primer sensitivity test shall be conducted on empty primed cases. The primed cases shall be obtained by disassembly of the sample cartridges. The disassembly shall be accomplished in such a manner as to cause the least possible distortion of the cartridge case. The test shall be conducted in accordance with TECP 700-700, Vol. III.

## 5. PREPARATION FOR DELIVERY

5.1 Packing.-Level A (Worldwide shipment) The cartridges shall be packed in accordance with Drawing F7553763 or as required by contract.

5.2 Marking and labeling.-Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

## 6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Applicable cartridge drawing number (see 3.1)

6.1.3 Type and level of packing.

6.1.4 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.5 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.6 Requirements for contractor to provide and maintain an inspection system in accordance with MIL-C-45208, Inspection System Requirements.



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6.2 Process deviation.-A process deviation is defined as a change in the initial approved basic method of manufacture. This also includes any operational changes, i.e., change in sub-contractor(s), changes which may alter the metallurgical or physical properties of the item which was approved initially in the basic method of manufacture. Any manufacturing deviations, after the "First Article Sample" has been approved, shall be submitted to the technical agency for approval.

6.3 Obturating Cups (Wax filled).-Prior to use, each obturating cup shall be filled with wax. The proof technician shall fill the cup in such manner that excess wax extends over the lip of the cup, the cup is then placed on a smooth surface mouth end down, and rubbed back and forth until the wax is flush with the lips of the cup. Overfilled or underfilled cups shall not be used. Obturating cups (wax filled) shall not be lubricated prior to use. Wax ingredients and method of preparation are as follows:

a. Wax ingredients

| | |
|---|---|
| Beeswax | 234.0 Grams |
| Paraffin | 6.0 Grams |
| Vaseline | 6.0 Grams |
| Castor Oil | 14.4 Grams |
| Lead Oxide (Red Ph O ) | 72.0 Grams |
| Iron Oxide (Fe O ) | 24.0 Grams |
| Rosin | 5% by volume |

b. Preparation.-The ingredients are weighed out in a casserole and heated on a steam bath until the waxes are melted. The mixture is then removed and stirred vigorously until slightly warm. The wax is then rolled out on a flat surface into sticks.

Custodian:

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Preparing activity:

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# APPENDIX "A"

## BULLET EXTRACTION TEST PROCEDURE

10.0 PURPOSE

The Bullet Extraction Test determines the force required to pull a bullet from its cartridge case. It is used as a measure of the uniformity and efficiency in which the bullets are secured in the cases.

20.0 EQUIPMENT

Equipment listed in the Bullet Extraction section of the appropriate Inspection Equipment List shall be used.

30.0 TEST PROCEDURE

30.1 Preparation for test

30.1.1 The testing machine should be calibrated either weekly or prior to each occasion of use, whichever is less frequent. Calibration points shall include at least 30 pounds, 60 pounds, 100 pounds, 150 pounds and 250 pounds. If calibration errors exceed three pounds at scale readings below 200 pounds or six pounds at scale readings above 200 pounds, then the necessary corrective action shall be taken to reduce the errors below these limits before the bullet extraction test is conducted. The method of calibration shall be by calibrated proof rings or calibrated spring balances which shall yield values within $\pm$ 1% of true values.

30.1.2 Each test cartridge shall be secured in the jig fixture and, using a standard drill press with a 1/8" drill, a hole shall be drilled completely through the bullet.

30.1.3 The number of cartridges as prescribed by the applicable specification are placed in a recessed holding block which shall be placed at a point convenient to the technician.

30.2 Conducting the Test

30.2.1 The drilled cartridge shall be positioned in a vertical plane into the test fixture and secured into the case-holding block. The hardened pin shall be positioned through the upper fixture and the bullet of the test cartridge.

30.2.2 The load shall be applied.



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APPENDIX "A" (CONT'D)

30.2.5 The case with the propellant should be placed upright in a recessed holding block and the extracted bullet placed point down in the mouth of the case. If the components are not needed, they shall be scrapped in accordance with local safety requirements.

40.0 RECORDING OF DATA

40.1 Test sheets shall show the following:

a. Force required to extract each bullet.

b. Grand average of the individual values

c. Minimum value

d. Maximum value

e. Testing Machine data



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## APPENDIX "B"

## RESIDUAL STRESS (MERCUROUS NITRATE) TEST PROCEDURE

10.0 PURPOSE

Mercurous Nitrate test is a visual means of determining if stresses exist in brass cartridge cases that may cause the cases to crack or split under conditions of storage or during service usage.

20.0 EQUIPMENT

Equipment listed in Mercurous Nitrate section of the appropriate Inspection Equipment List shall be used.

30.0 MANDATORY SAFETY REQUIREMENTS

30.1 Food shall not be stored or eaten in vicinity in which these tests are conducted.

30.2 Acid resistant apron and gloves or the equivalent shall be worn by each test technician.

30.3 Face shields shall be worn at all times during the pouring or mixing of acids and water.

30.4 Asbestos or heat insulating gloves shall be worn during the heat volatilization phase of the test to facilitate handling.

30.5 During the entire period of volatilization, the oven door shall be closed.

30.6 Extreme care shall be exercised in the mixing of acid with water; this shall be accomplished by pouring the acid into the water.

30.7 Test shall be conducted under a canopy or hood having a forced draft ventilation system to remove the noxious fumes. The disposition of tested components shall be governed by local regulations.

40.0 TEST PROCEDURE

40.1 Preparation of Solution

40.1.1 Nitric-Acid Solution: Four hundred (400) milliliters of nitric acid (of specific gravity 1.42) are dissolved in five hundred (500) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter. (The resultant specific gravity will be 1.25).



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APPENDIX 'B' (CONT'D)

40.1.2 Mercurous Nitrate Solution: Ten (10) grams of mercurous nitrate and ten (10) milliliters of nitric acid (of specific gravity 1.42) are dissolved in four hundred (400) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter.

40.2 Test of Cartridges

40.2.1 The cartridges are positioned in both nitric acid and mercurous nitrate solutions in a vertical position, with the head of the case down. The depth of the solution is adjusted until it completely covers the mouth of the case but does not cover more than one-half the body of the bullet.

40.2.2 One-half of the number of cartridges prescribed in the applicable specification for the Mercurous Nitrate test is submerged in the nitric acid solution. After thirty (30) seconds the cartridges are withdrawn, rinsed in water and the excess water removed. The cartridges are then submerged for fifteen (15) minutes in the mercurous nitrate solution. Immediately upon removal, the surface of the cartridge case is examined under a magnification (10 to 15 diameters) for splits and cracks. All splits and cracks shall be reported.

40.3 Test of Cartridge Cases

40.3.1 The cases for this test are obtained by disassembly of cartridges. The primers must be fired prior to using the cases.

40.3.2 The cases are positioned in both nitric acid and mercurous nitrate solutions in a vertical position, with the head of the case down. The depth of each solution is adjusted until it completely covers the mouth of the case but does not rise more than one-half inch above the mouth of the case.

40.3.3 Cases from the remaining half of the number of cartridges prescribed in the applicable specification for the Mercurous Nitrate Test (after the bullet and propellant have been removed and primer fired) are submerged in nitric acid solution. After thirty (30) seconds, the cases are withdrawn, rinsed in water and the excess water removed. The cases are then submerged for fifteen (15) minutes in the mercurous nitrate solution. Immediately upon removal, the mercury on the surface of the case is volatilized by the application of heat and the surface then examined under magnification (10 to 15 diameters) for splits and cracks. All splits and cracks shall be reported.



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APPENDIX "B" (CONT'D)

NOTE: (a) A split is defined as a separation of the metal entirely through the wall of the case.

(b) A crack is a surface condition and represents a separation of the metal partially through the wall of the case. Cracks are not considered to be splits.

40.3.4 A suggested method for determining splits and cracks follows:

Splits in the cartridge case, after the mercurous nitrate test are detected by filling the case with water (water temperature 70°F to 100°F) to the mouth until a convex miniscus condition exists and placing the thumb of the hand over the mouth of the case. If the case is split, this exerts sufficient pressure to force the water through the opening.

Splits in the assembled cartridge are not so easily detected unless the split is at the mouth of the case and the bullet metal is seen through the split. If the split is in the body, the only way to determine whether it is a split or crack is to disassemble and follow the procedure used for the case check.

50.0 RECORDING OF DATA

50.1 Results of the test shall be recorded.

50.1.1 All splits and cracks shall be reported.



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## APPENDIX "C"

## VELOCITY TEST PROCEDURE

10.0 PURPOSE

The Velocity Test is a precisely controlled test to determine the velocity uniformity and level of the ammunition lot under test.

20.0 EQUIPMENT

20.1 Equipment listed in the Velocity section of the appropriate Inspection Equipment List shall be used.

20.2 The firing range shall be arranged as shown on Figure C-1.

20.3 Pier or mount of solid construction on which the test fixture assembly shall be mounted.

30.0 USE OF STANDARD (REFERENCE) CARTRIDGES

30.1 Standard (reference) cartridges shall be used to establish range and equipment corrections prior to firing an ammunition lot for acceptance. One standard (reference) cartridge shall be fired for each cartridge of the test sample to be fired, up to twenty (20) cartridges. When the test consists of more than twenty (20), but not more than forty (40) cartridges, then twenty (20) standard (reference) cartridges shall be fired. When the test consists of more than forty (40) cartridges, then one standard (reference) cartridge shall be fired for every two (2) test cartridges.

30.2 After the required number of standard (reference) cartridges has been fired, the actual average velocity of the standard (reference) cartridges shall be compared with the assessed value. If the assessed value is higher than the actual average velocity of the standard (reference) cartridges, the difference is a plus correction and shall be added to the average velocity of the test cartridges. If the assessed value is lower than the actual average velocity of the standard (reference) cartridges, the difference is a minus correction and shall be subtracted from the average velocity of the test cartridges. If the assessed value and the actual average velocity of the standard (reference) cartridges are identical, then no correction is applied.

30.3 The velocity-barrel assembly will be acceptable provided the results obtained with the standard (reference) cartridges are within $\pm$ 35 feet-per-second of the assessed value.



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APPENDIX "C" (CONT'D)

30.4 It is suggested that control charts shall be maintained, for record purposes, of the results from each barrel in which standard (reference) cartridges are fired. Every correction obtained shall be compared with the previous ones to determine whether the test conditions remain stable to a certain degree.

40.0 TEST PROCEDURE

40.1 Pre-firing (Preparation for test)

40.1.1 The required number of test cartridges and standard (reference) cartridges shall be placed in a vertical position, primer-end down, in recessed holding blocks. The cartridges shall be permitted to come to a temperature of $60^{\circ}$ to $80^{\circ}$F. prior to being placed in the temperature-controlled room or container. The recessed holding blocks containing the two series of cartridges shall be placed in a temperature-controlled room or container in such a manner that all cartridges are subjected to a uniform temperature for a minimum of two hours, prior to firing. The container or room shall be maintained at a temperature of $70^{\circ}$F., $\pm$ $2^{\circ}$F., with a relative humidity of 60 percent, $\pm$ 5 percent and be of sufficient capacity to allow free circulation of air.

40.1.2 The velocity-barrel assembly is assembled in the test fixture, on the mount. The chamber and bore of the barrel are wiped dry. The barrel is then boresighted into position.

40.1.3 The velocity-barrel assembly shall be in accordance with the applicable drawings and the following measurements shall be made before it is placed in service, after misfires, pierced primers, flowbacks, or whenever a part replacement has been made in the breech block assembly.

| | Limits |
|---|---|
| Firing-pin protrusion | .060" - .068" |
| Firing-pin indent | .011" - .015" |
| Headspace | .058" - .060" |

40.1.4 Lumiline screens are checked for position. It is of the utmost importance that the lumiline screens be placed in their proper positions, measurements must be accurate.

40.2 During Firing

40.2.1 Five (5) warmer (fouling) shots shall be fired. Velocity readings should be recorded by the chronographer to assure that the velocity-measuring equipment is functioning properly.



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APPENDIX "C" (CONT'D)

40.2.2 The recessed holding block containing the required number of standard (reference) cartridges is removed from the controlled temperature room or container and placed at a point convenient to the technician, provided the temperature of the firing room is 70°F., ± 5°F., otherwise the cartridges shall be placed in an insulated box (five cartridges at a time) which has been conditioned at 70°F., ± 5°F., and the box placed at a point convenient to the technician, the cartridges are then removed singly from the insulated box immediately before firing. If an insulated box is not available, then the cartridges shall be removed singly from the controlled-temperature room or container immediately before firing.

40.2.3 In order that the propellant shall be uniformly positioned from cartridge to cartridge before firing, attention to detail is necessary in handling and chambering the cartridge. The cartridge shall be first held vertically, bullet upward, and then rotated slowly, end over end in a vertical plane, stopping the rotation momentarily after 180° of rotation when the bullet is downward, and then continuing through the remainder of 360°, stopping with the cartridge again bullet-end upward. The bullet-end of the cartridge should now be lowered slowly to a position slightly above horizontal. The cartridge shall be chambered very carefully, taking care that the primer-end of the case is not elevated above the bullet-end. (The object is to have the cartridge seated in the chamber ready to fire, with the propellant in a loose condition at the primer-end of the case.)

40.2.4 The breech-block shall be closed gently and the trip lever to which the lanyard is attached, shall be carefully engaged to the hammer. If the technician encounters any difficulty closing the breech-block or engaging the trip lever, the test shall be discontinued until such difficulty is corrected. If any delay should occur after the cartridge is placed in the chamber, and the duration of the delay is approximately 30 seconds or longer, that cartridge shall be extracted and another inserted in its place.

40.2.5 The technician retires to a safe position and pulls the lanyard with a smooth firm motion. The velocity of the shot shall be recorded by the chronographer. The breech-block shall be opened, the fired case extracted and visually examined by the technician for possible case casualties.

40.2.6 The procedure prescribed in 40.2.3 through 40.2.5 is repeated until the required number of standard (reference) cartridges have been fired.

40.2.7 The velocity correction shall then be obtained as prescribed in 30.2.

40.2.8 Continuous air cooling should be used on the barrel. If air cooling is not available, firing should be regulated so that one cartridge is fired every 30 seconds. Barrel shall be allowed to cool to ambient temperature between each



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series of tests, or after a maximum number of thirty (30) cartridges has been fired. At no time shall the exposed metal surface of the test barrel become too hot to grasp with the bare hands (approximately 140°F.).

40.2.9 If the average velocity of the standard (reference) cartridges is not within ± 35 F/S of the official assessed velocity value, the test barrel shall be removed from the test and another test barrel substituted and another series of standard (reference) cartridges fired. If this firing fails to produce satisfactory velocity results, the cause thereof shall be identified and eliminated before resuming the test.

40.2.10 After the barrel has cooled to ambient temperature, warming (fouling) shots shall be fired in accordance with 40.2.1.

40.2.11 The test cartridges shall then be fired following the procedure prescribed in 40.2.2 through 40.2.6.

40.2.12 The velocity correction obtained with the standard (reference) cartridges shall then be applied to the mean velocity obtained with the test cartridges as prescribed in 30.2.

40.2.13 If any of the personnel conducting the test observes any abnormality that may invalidate the velocity measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority. If any result appears abnormal, even though the possible cause cannot be immediately assigned, the circumstances shall be reported immediately for investigation and decision by proper supervisory authority.

50.0 RECORDING OF DATA

50.1 Results of both, standard (reference) and test cartridges, should be recorded directly on the test sheet form.

50.1.1 Standard (reference) Cartridges: The individual velocities, average velocity and extreme variation of the individual velocities.

50.1.2 Test Cartridges: The individual velocities, average velocity (not corrected), velocity correction, average velocity (corrected), extreme variation and standard deviation of the individual velocities.

50.1.3 Number and type of case casualties.



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APPENDIX "C" (CONT'D)

50.2 The following test weapon data shall be recorded on the test sheets.

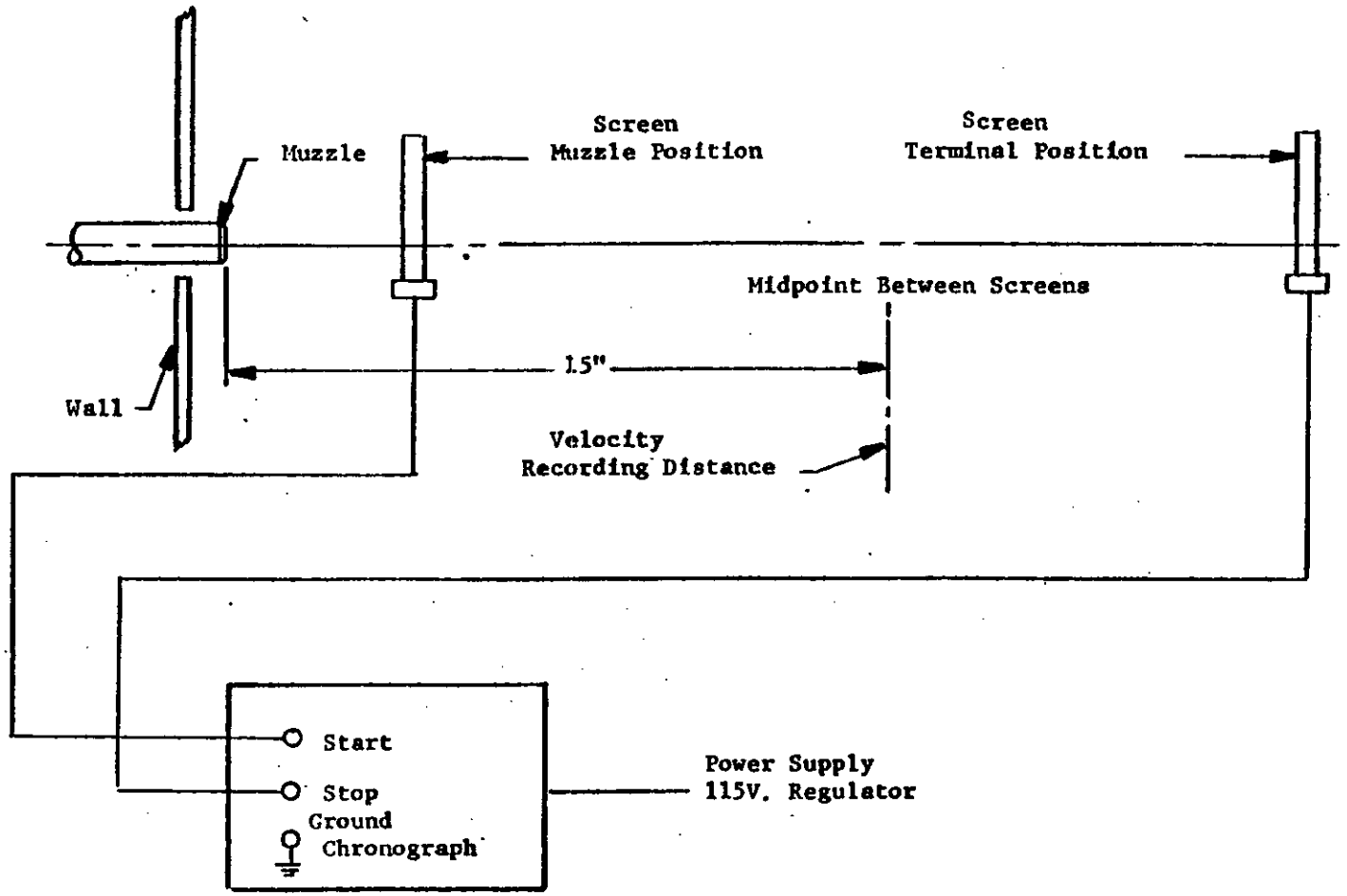
a. Receiver number

b. Barrel number

c. Total number of cartridges fired in barrel



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Range Equipment, Velocity Test for Bullet Type, Cal. .38



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# APPENDIX "D"

## CHAMBER-PRESSURE TEST PROCEDURE

10.0 PURPOSE

The Chamber-Pressure test is a precisely controlled test to determine the chamber-pressure exerted, expressed in pounds-per-square-inch, in the chamber of the gun. The test is performed as a safety measure to insure that the pressure developed by the ammunition is safe for firing in the weapon for which it is intended.

20.0 EQUIPMENT

20.1 Equipment listed in the Chamber-Pressure section of the appropriate Inspection Equipment List shall be used.

20.2 The firing range shall be arranged as shown in Figure D-1.

20.3 Pier or mount of solid construction on which test fixture assembly shall be mounted.

30.0 USE OF STANDARD (REFERENCE) CARTRIDGES

30.1 Standard (reference) cartridges shall be used to establish range and equipment corrections prior to firing an ammunition lot for acceptance. One standard (reference) cartridge shall be fired for each cartridge of the test sample to be fired, up to twenty (20) cartridges. When the test consists of more than twenty (20), but not more than forty (40) cartridges, then twenty (20) standard (reference) cartridges shall be fired. When the test consists of more than forty (40) cartridges, one (1) standard (reference) cartridge shall be fired for every two (2) test cartridges.

30.2 After the required number of standard (reference) cartridges has been fired, the actual average chamber-pressure of the standard (reference) cartridges shall be compared with the assessed value. If the assessed value is higher than the actual average chamber-pressure of the standard (reference) cartridges, the difference is a plus correction and shall be added to the average chamber-pressure of the test cartridges. If the assessed value is lower than the actual average chamber-pressure of the standard (reference) cartridges, the difference is a minus correction and shall be subtracted from the average chamber-pressure of the test cartridges. If the assessed value and the actual chamber-pressure of the standard (reference) cartridges are identical, then no correction is applied.



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APPENDIX "D" (CONT'D)

30.3 The pressure-barrel assembly will be acceptable provided the results obtained with the standard (reference) cartridges are within ± 3500 PSI of the assessed value.

30.4 It is suggested that control charts be maintained, for record purposes, of the results from each barrel in which standard (reference) cartridges are fired. Every correction obtained shall be compared with the previous ones to determine whether the test conditions remain stable to a certain degree.

40.0 MEASUREMENT OF COPPER PRESSURE CYLINDERS

40.1 The chamber-pressure shall be determined by using the radial copper pressure cylinders. One cylinder shall be used for each cartridge fired.

40.1.1 The copper pressure cylinders shall be measured individually using a properly calibrated micrometer graduated to 0.0001 inch. The cylinders shall be measured prior to firing, measurements recorded and the cylinders placed in a recessed holding block in such a manner that their identity is maintained throughout the test.

40.1.2 Upon completion of the firing, the cylinders shall be measured again and the decrease in length obtained for each cylinder, by subtraction of the actual readings. The decrease in length obtained for each cylinder shall then be applied to the appropriate tarage table (a tarage table is supplied with each box of copper pressure cylinders) and the corresponding chamber-pressure (PSI) shall be recorded on the test report form in such a manner that the velocity of each cartridge can be identified with the corresponding chamber pressure obtained.

40.2 Whenever a new supply of copper pressure cylinders is received involving the use of different tarage tables, they shall be tested in comparison with the cylinders being replaced by firing forty-four (44) standard (reference) cartridges for chamber pressure in a barrel assembly of known characteristics, using twenty-two (22) of the new cylinders in alternation with twenty-two (22) of the old cylinders. The mean chamber pressure obtained with the new cylinders shall not vary by more than ± 800 PSI from the mean obtained with the old cylinders. If this limit is exceeded, the new lot is considered unacceptable and reported to the responsible agency.

40.3 TEST PROCEDURE

40.3.1 Pre-firing (Preparation for test)

40.3.1.1 The required number of test and standard (reference) cartridges shall be placed in a vertical position primer-end down, in recessed holding blocks.



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APPENDIX "D" (CONT'D)

The cartridges shall be permitted to come to a temperature of 60° to 80°F. prior to being placed in the temperature-controlled room or container . The recessed holding blocks containing the two series of cartridges shall be placed in a temperature-controlled room or container in such a manner that all cartridges are subjected to a uniform temperature for a minimum of two hours, prior to firing. The container or room shall be maintained at a temperature of 70°F., ± 2°F., with a relative humidity of 60 percent, ± 5 percent, and be of sufficient capacity to allow free circulation of air.

40.3.1.2 The pressure-barrel assembly is assembled in the test fixture on the mount. The chamber and bore of the barrel are wiped dry. The barrel is then boresighted into position.

40.3.1.3 The following measurements shall be made before a barrel assembly is placed in service, after misfires, pierced primers, flowbacks, or whenever a part replacement has been made in the breech-block assembly.

| | Limits |
|---|---|
| Firing-pin-indent | .011" - .015" |
| Firing-pin protrusion | .060" - .068" |
| Headspace | .058" - .060" |

40.3.1.4 Lumiline screens are checked for position. It is of utmost importance that the lumiline screens be placed in their proper positions; measurements must be accurate.

40.3.1.5 The required number of copper pressure cylinders are measured as prescribed in 40.1.1.

40.3.1.6 The recessed holding block containing the individually measured copper pressure cylinders, a sufficient quantity of obturating cups and the pressure piston previously fitted to the barrel assembly should be placed at a point convenient to the technician.

40.3.1.6.1 The obturating cups (wax filled) are not changed after firing each shot. It is permissible to fire a maximum of twenty-two (22) cartridges with the same obturating cup. The first two cartridges are fired to properly seat the obturating cup in the piston hole, the results obtained with these two cartridges are recorded for information only and are not considered in the final averages.



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APPENDIX "D" (CONT'D)

40.3.2.1 Five (5) warmer (fouling) shots shall be fired. To fire the warming (fouling) shots, it shall be necessary to service the chamber-pressure barrel assembly with an obturating cup, the pressure piston, and a copper pressure cylinder. The same cylinder and obturating cup may be used when all the warming shots are being fired, the anvil is screwed down on the cylinder following each shot. Velocity readings should be recorded by the chronographer to assure that the velocity-measuring equipment is functioning properly.

40.3.2.2 The recessed holding block containing the standard (reference) cartridges is removed from the controlled-temperature room or container and placed at a point convenient to the technician, provided the temperature of the firing room is 70°F., ± 5°F., otherwise the cartridges shall be placed in an insulated box which has been conditioned at 70°F., ± 5°F., and the box placed at a point convenient to the technician; the cartridges are then removed singly from the insulated box immediately before firing. If an insulated box is not available, then the cartridges shall be removed singly from the controlled-temperature room or container immediately before firing.

40.3.2.3 The dummy cartridge is placed in the chamber of the barrel assembly. The obturating cup shall be placed in the piston hole, mouth-end down, and partially seated using the stem of the knockout tool. The shank of the pressure piston shall be dipped in oil, SAE40 or equivalent, and the oil then be allowed to drain from the piston. The drop of oil adhering to the bottom of the piston shank shall then be removed by touching the bottom of the piston shank to a cleaning patch. The piston shall then be inserted into the piston hole and pressed down on the obturating cup until the piston has reached its correct final position. The head of the piston and the bottom of the anvil shall be wiped dry and free of oil. The copper cylinder shall be put in place and centered between the head of the piston and the bottom of the anvil. The anvil shall be screwed down lightly on the cylinder, using the thumb and forefinger, but not under stress. The dummy cartridge shall then be removed from the chamber. Chamber and bore are examined for possible obstructions.

40.3.2.4 In order that the propellant shall be uniformly positioned from cartridge to cartridge, attention to detail is necessary in handling and chambering the cartridge. The cartridge shall first be held vertically, bullet upward, and then rotated slowly, end over end in a vertical plane, stopping the rotation momentarily after 180° of rotation when the bullet is downward, and then continuing through the remainder of 360°, stopping with the cartridge again bullet-end upward. The bullet-end of the cartridge should now be lowered slowly to a position slightly above horizontal. The cartridge shall be chambered very carefully, taking care that the primer-end of the case is not elevated above the bullet-end. (The object is to have the cartridge seated in the chamber ready to fire, with the propellant in a loose condition at the primer-end of the case.)



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APPENDIX "D" (CONT'D)

40.3.2.5 The breech-block shall be closed gently. If the technician encounters any difficulty closing the breech-block or engaging the trip lever, the test shall be discontinued until such difficulty is corrected. If any delay should occur after the cartridge is placed in the chamber and the duration of the delay is approximately 30 seconds or longer, that cartridge shall be extracted and another inserted in its place.

40.3.2.6 The technician makes a final check to assure that the anvil is screwed to a snug position on the copper cylinder. Care is taken to see that the copper cylinder is not compressed by the anvil prior to firing. The proper torque to be applied to the thumbscrew is about one pound-inch. (This torque can be estimated with satisfactory accuracy by an experienced technician; to familiarize inexperienced technicians with the desired degree of tightness; a torque measuring device may be employed. This can be accomplished by drilling and tapping (threading) an axial hole in the knurled head of the anvil (thumb-screw) and inserting a bolt or screw to which a torque-measuring wrench can be attached. The usefulness of the anvil (thumb-screw) is not impaired by this modification.)

40.3.2.7 The trip lever, to which the lanyard is attached, shall be engaged gently to the hammer. The technician retires to a safe position and pulls the lanyard with a smooth firm motion. The velocity of the shot shall be recorded by the chronographer.

40.3.2.8 The breech-block shall be opened, the fired case extracted and visually examined by the technician for possible case casualties. The copper cylinder shall be removed and placed in the recessed holding block. (Because the same obturating cup is used to fire twenty-two (22) cartridges, it is not necessary to remove either the obturating cup or the piston from the piston hole from shot to shot). The piston is then pressed down in the piston hole to its correct final position; this can be accomplished by gently tapping the head of the piston with the knockout tool or by placing a copper cylinder on the head of the piston and screwing the anvil down on it forcing the piston into its correct final position. The cylinder used for this purpose is then removed from the head of the piston and replaced with the next cylinder in the recessed holding block. The anvil shall be screwed down lightly on the cylinder, using the thumb and forefinger, but not under stress. The chamber and bore of the test weapon are checked for obstructions.

40.3.2.9 The procedure prescribed in 40.3.2.4 through 40.3.2.8 is repeated until the twenty-two (22) cartridges have been fired.

40.3.2.10 The piston is then removed from the piston hole. The knockout tool is then used to force the obturating cup from the piston hole into the chamber. The obturating cup is then removed from the chamber using a cleaning rod with a dry cleaning patch attached.



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APPENDIX "D" (CONT'D)

40.3.2.11 Continuous air cooling should be used on the barrel; however, if air cooling is not available, firing should be regulated so that one cartridge is fired every thirty (30) seconds. At no time shall the exposed metal surface of the test barrel become too hot to grasp with the bare hands (approximately 140°F).

40.3.2.12 The copper pressure cylinders whose identities are maintained throughout the test are measured in the same manner as they were prior to the test. The difference in length (set in inches) is then applied to the proper tarage table, and the corresponding PSI is obtained and recorded on the report form. The chamber pressure correction shall then be obtained as prescribed in 30.2.

40.3.2.13 If the average chamber pressure of the standard (reference) cartridges is not within ± 3500 PSI of the official assessed value, the test barrel shall be removed from the test, another test barrel substituted and another series of standard (reference) cartridges fired. If this firing fails to produce satisfactory results, the cause thereof shall be identified and eliminated before resuming the test.

40.3.2.14 After the barrel has cooled to ambient temperature, warming (fouling) shots shall be fired in accordance with 40.3.2.1.

40.3.2.15 The test cartridges shall then be fired following the procedure prescribed in 40.3.2.2 through 40.3.2.9.

40.3.2.16 The copper cylinders used with the test cartridges are measured in the same manner as they were prior to the test. The chamber-pressure readings shall be placed on the test sheet in such a manner that the velocity and chamber-pressure of each shot is identifiable. The chamber-pressure correction obtained with the reference cartridges shall be applied to the average chamber-pressure of the test cartridges, as prescribed in 30.2.

40.3.2.17 If any of the personnel conducting the test observes any abnormality that may invalidate the velocity or pressure measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority. If any result appears abnormal, even though the possible cause cannot be immediately assigned, the circumstances should be reported immediately for investigation and decision by proper supervisory authority.

50.0 RECORDING OF DATA

50.1 Results of both standard (reference) and test cartridges should be recorded directly on the test sheet form. Velocities shall be recorded to the nearest FPS: chamber-pressures shall be recorded to the nearest 100 PSI.



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APPENDIX "D" (CONT'D)

50.1.1 Standard (reference) cartridges.-The individual velocities and chamber-pressures, average velocity and chamber-pressure, and extreme variations of velocity and chamber-pressure shall be recorded.

50.1.2 Test cartridges.-The individual velocities and chamber pressures, average velocity and chamber pressure (not corrected), chamber pressure correction, average chamber pressure (corrected), extreme variation of the individual velocities and the individual chamber pressures shall be recorded.

50.1.3 Number and type of case casualties.

50.2 The following test weapon data shall be recorded on the test sheet:

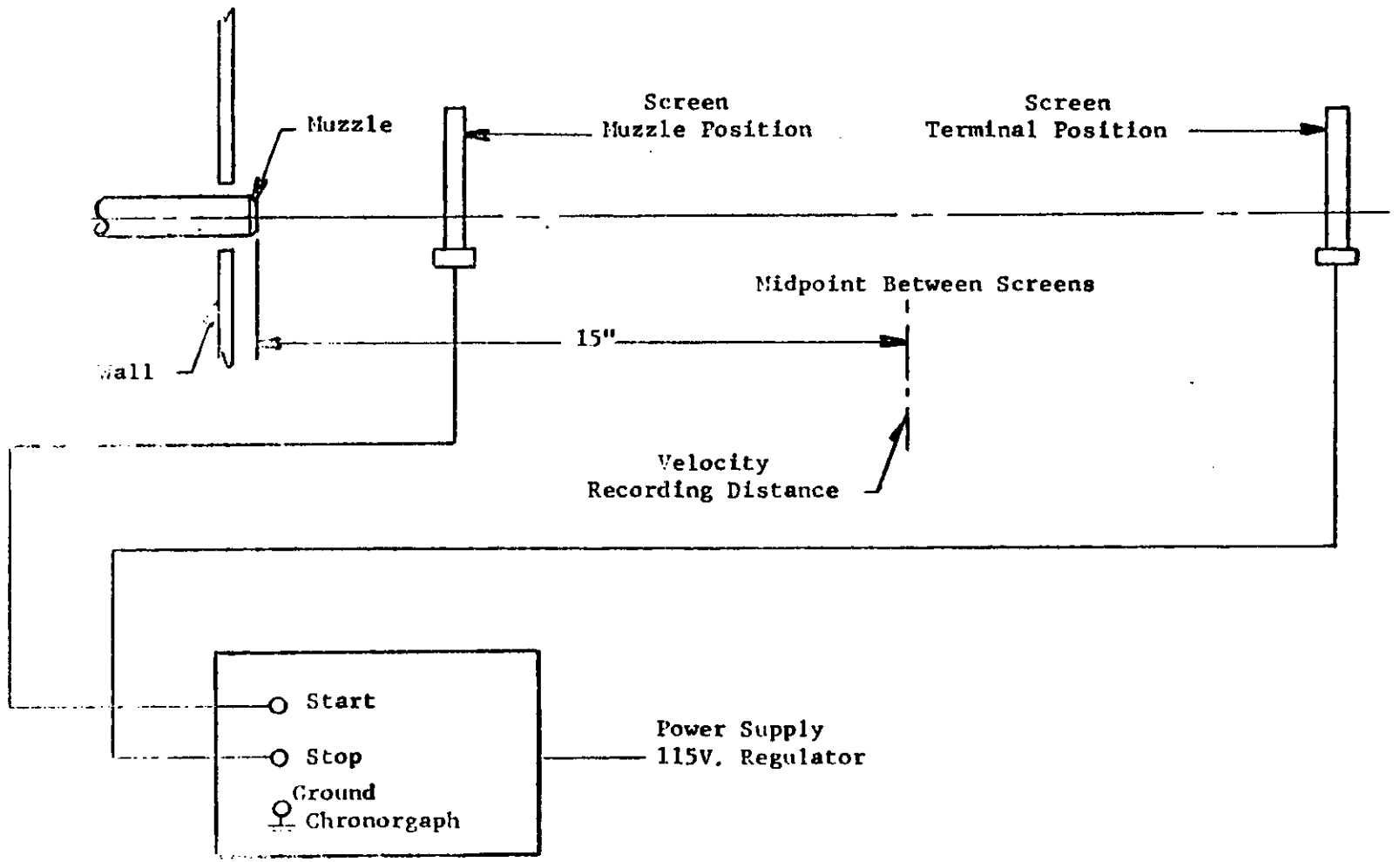
a. Receiver number

b. Barrel number

c. Total number of cartridges fired in barrel



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Range Equipment, Velocity Test for Bullet Type, Cal. .38



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## APPENDIX "E"

## ACCURACY TEST PROCEDURE

10.0 PURPOSE

The Accuracy Test is fired under controlled conditions at a target located at a specified distance from the weapon to determine the uniformity and dispersion of bullets. In order to minimize weapon inaccuracies, the test shall be fired in a rigidly mounted test weapon.

20.0 EQUIPMENT

20.1 Equipment listed in the Accuracy section of the appropriate Inspection Equipment List shall be used.

20.2 The target size for Accuracy tests shall not be less than 24" x 24".

20.3 The pier or mount on which the test assembly is mounted shall be of solid construction.

30.0 TEST PROCEDURE

30.1 Pre-firing (Preparation for test)

30.1.1 The accuracy test weapon is assembled and locked in position on the mount and boresighted into position. It is of prime importance that the assembly be mounted properly so that the weapon maintains its original position from shot to shot.

30.1.2 The following measurements shall be made before a test weapon is placed in service, after misfires, pierced primers or flowbacks; or whenever a part replacement has been made in the assembly:

| | Limits |
|---|---|
| Firing-pin protrusion | .060" - .068" |
| Firing-pin indent | .011" - .015" |
| Headspace | .058" - .060" |

30.1.3 The test cartridges shall be placed at a point convenient to the technician. It is not necessary to condition the ammunition at a specified temperature, prior to firing.

30.1.4 At least two (2) but preferably three (3) weapons shall be used. The number of targets to be fired shall be divided among the number of weapons to be used.



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APPENDIX "E" (CONT'D)

30.2 During Firing

30.2.1 A sufficient number of unrecorded cartridges of the type of ammunition under test shall be fired to assure that the test weapon is correctly sighted on the target, but in any event, a minimum of three (3) cartridges shall be fired to warm and foul the weapon when it is first put into service and after it has been cleaned or cooled. The technician observes the location of the shots on the target so that the alignment of the weapon may be adjusted, if necessary.

30.2.2 After the warming (fouling) cartridges have been fired, the target shall be changed so as to present a fresh surface for the succeeding target. Thereafter, the target shall be changed after each target of ten cartridges has been fired.

30.2.3 Consideration is not given to the position of the propellant in the cartridge case, except that the manner of handling the ammunition from cartridge to cartridge is reasonably uniform. Ten (10) cartridges shall then be fired in even sequence at a rythmic uniform rate, as rapidly as service of the weapon permits.

30.2.4 The approximate location of the center of impact and the estimated size (Extreme Vertical and Extreme Horizontal) of the target first fired shall be observed by the technician in order that alignment of the weapon may be adjusted, if necessary.

30.2.5 The procedure prescribed in 30.2.3 and 30.2.4 shall then be repeated until the specified number of targets have been fired in the first weapon, five (5) targets maximum, or when the exposed metal surface of the test barrel becomes too hot to grasp with the bare hands (approximately 140 degrees F)., at which time the barrel shall be cooled to ambient temperature before the test is continued.

30.2.6 After the required number of targets has been fired with the first weapon, that weapon shall be removed from the mount and cooled to ambient temperature.

30.2.7 The second test weapon shall then be assembled on the mount and the procedure prescribed in 30.2.1 and 30.2.6 is repeated. If a third test weapon be used, the same procedure shall be repeated.

30.2.8 If it is necessary to use a weapon after it has been cooled, the chamber and bore shall be cleaned before any additional firing, and the three warming (fouling) cartridges shall be repeated.



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APPENDIX "E" (CONT'D)

30.2.9 The technician shall record all pertinent information relative to the test, i.e., misfires, failue to trace, case casualties, etc. Whenever the validitv of the test is in doubt, test equipment can be checked by firing ammunition of known accuracv.

30.3 MEASUREMENT OF TARGET

30.3.1 Explanation of Terms

30.3.1.1 Diagonal (D) - After drawing a rectangle whose four lines pass through the outermost shot-holes on a target, the diagonal (hypotenuse) is measured to the nearest tenth of an inch.

30.3.1.2 Mean Diagonal (M.D.) - The arithmetic mean of all diagonals.

30.3.1.3 Figure of Merit (F.M.) - The arithmetic average of the Extreme Vertical and the Extreme Horizontal.

30.3.1.4 Mean Radius (M.R.) - The arithmetic mean of the distances between the centers of all shot-holes and a point of the target called the Center of Impact.

30.3.1.5 Extreme Vertical (Ex. Ver.) - Vertical distance between the center of the hole made by the uppermost shot and the center of the hole made b y the lowermost shot.

30.3.1.6 Extreme Horizontal (Ex. Hor.) - Horizontal distance between the center of the hole made by the shot farthest to the right and the center of the hole made by the shot farthest to the left.

30.3.1.7 Extreme Spread (Ex. Spr.) - Distance between centers of the shot-holes farthest apart.

30.3.1.8 Center of Impact (C.I.) - Defined as the point at which the algebraic sum of the components of the distances to the center of each shot-hole is zero.

30.3.2 The procedure for obtaining the Diagonal (D) for each target shall be as follows:

30.3.2.1 On each target, draw a rectangle whose four sides pass through the centers of the outermost holes (See Figure E-3). The top and bottom lines of the rectangle shall be horizontal; the vertical lines shall be perpendicular to the horizontal lines.



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APPENDIX "E" (CONT'D)

30.3.2.2 The Diagonal (D) of each rectangle shall then be measured to the nearest tenth of an inch. The Diagonal may be measured from the upper left corner to the lower right corner, or from the upper right corner to the lower left corner.

30.3.2.3 The Mean Diagonal (M.D.) shall be obtained by adding the individual diagonal measurements and dividing by the number of targets.

30.3.3 Mean Radius for targets shall be determined in the manner indicated below, and illustrated by Figures E-1 and E-2.

30.3.3.1 Draw a horizontal line, PA, through the center of the lowest shot-hole on the target and extend it horizontally in both directions until its length is equal to or greater than the width of the group of shots comprising the target.

30.3.3.2 Draw a vertical line, PB, through the center of the shot-hole on the extreme left of the target perpendicular to and meeting line PA at P. Extend line PB above PA until the length of PB is equal to or greater than the height of the group of shots comprising the target.

30.3.3.3 Measure (by means of the scale or opisometer) the perpendicular distance Y from the center of each shot-hole on the target to line PA. Add the distances so obtained and divide by the number of shots. The result of this calculation is the average vertical distance of the shot-holes above the base line PA and is designated by Y. Similarly, measure the perpendicular distance X from the center of each shot-hole to line PB, add the distances so obtained and divide by the number of shots. The result is the average horizontal distance of the shot-holes from the vertical ordinate PB, and is designated X.

30.3.3.4 Location of Center of Impact - From P, measure along PB a distance, PD, equal to Y and along PA measure a distance, PE, equal to X. At points E and D erect the respective perpendiculars EC and DC to lines PA and PB. The intersection of these perpendiculars is the Center of Impact of the target, C. Steps 30.3.3.1 and 30.3.3.4 inclusive, are illustrated in Figures E-1.

30.3.3.5 Mean Radius (M.R.) - From the Center of Impact, measure the distance to the center of each shot-hole of the target. Add these distances and divide by the number of shot-holes, the result is the M.R. Figure E-2. illustrates this procedure.

30.3.3.6 If one or more shots miss the target in any ten shot series, the entire accuracy test shall be considered invalid and refired at a larger target, as necessary to insure hits.



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APPENDIX "E" (CONT'D)

40.0 RECORDING OF DATA

40.1 All test results shall be recorded to the nearest one-tenth of an inch.

40.2 Test sheets shall show the following:

a. Results of all specified requirements

b. Number of type of case casualties

c. Misfires

d. Stripped bullet

40.3 The following test weapon data shall be recorded on the test sheets:

a. Receiver number

b. Barrel number

c. Total number of cartridges fired in barrel

d. Headspace measurement

40.4 FACTORS AFFECTING ACCURACY TESTS

40.4.1 The external factors should be controlled as closely as possible in order to obtain results that are representative of the inherent accuracy of the ammunition.

40.4.2 The dimensions and condition of accuracy weapons, as well as the manner of placing the weapon in the rest, are of prime importance.

40.4.3 The technique of test can affect accuracy results. If reproducibility in the manner in which the cartridges are handled, chambered and fired be poor, a larger Mean Diagonal or Mean Radius is obtained than if the cartridges are tested in a uniform manner.

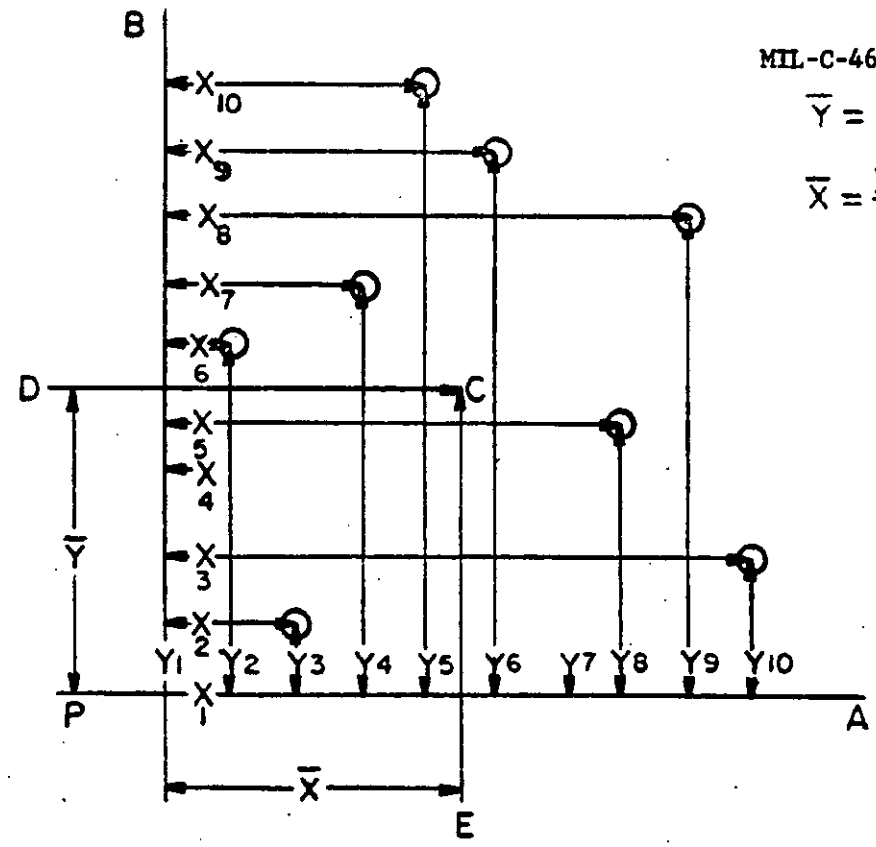
40.4.4 The temperature of the test barrel should be controlled so that the exposed metal surface of the barrel does not become too hot to grasp with the bare hands (approximately 140 degrees F), at which time the weapon shall be cooled to ambient temperature before the test is continued.



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$$\bar{Y} = \frac{\Sigma Y}{N} = PD$$

$$\bar{X} = \frac{\Sigma X}{N} = PE$$




$$M.R. = \frac{\Sigma Z}{N}$$

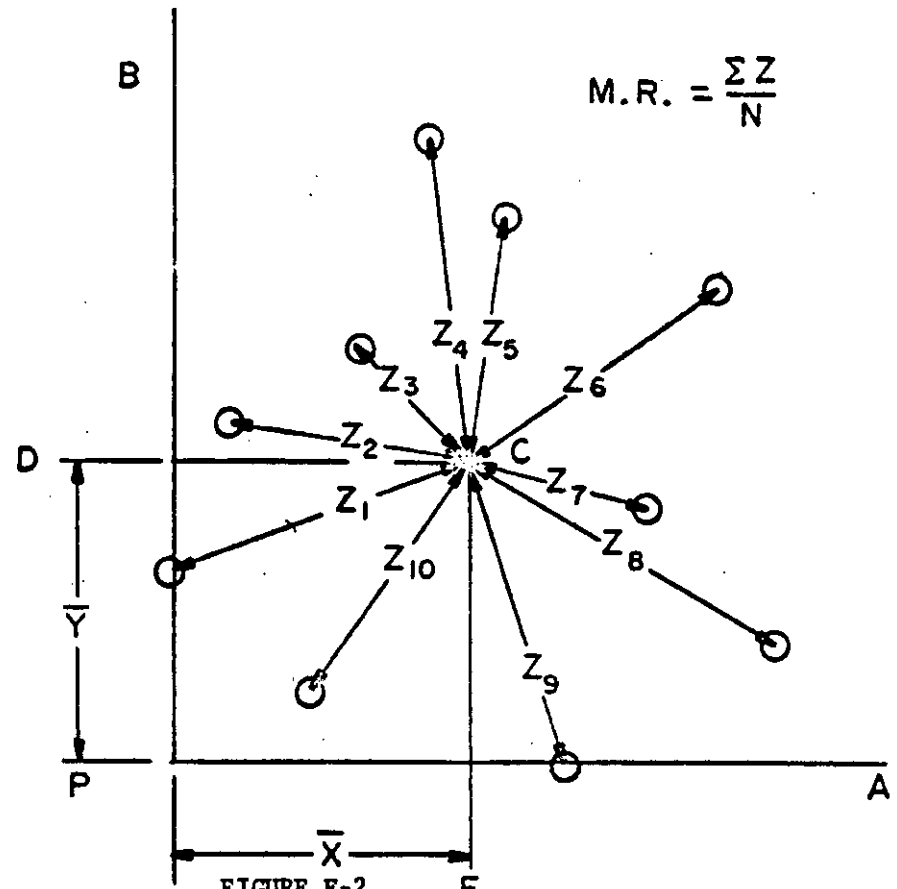



FIGURE E-2



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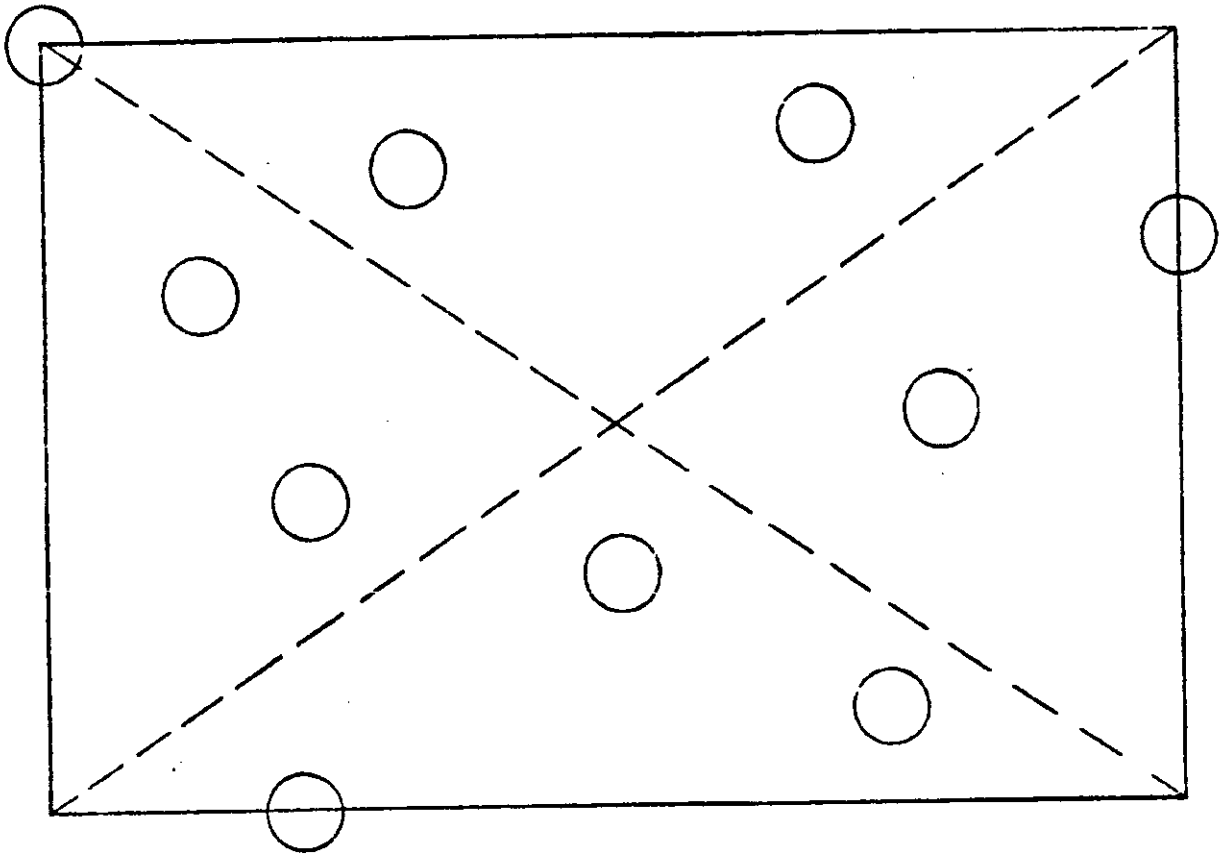


FIGURE E-3

MEAN DIAGONAL



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# APPENDIX "F"

## FUNCTION AND CASUALTY TEST PROCEDURE

10.0 PURPOSE

To determine if the ammunition can be expected to perform satisfactorily in the service weapons for which it has been designed under conditions of field usage. Ammunition may be ballistically satisfactory, i.e., it may comply with individual performance specifications such as velocity, pressure, etc., yet be unfit for use in the field because of undesirable characteristics which jeopardize the safety of personnel.

Casualties and malfunctions can be caused either by the ammunition or by the weapon in which it is fired, so that, to a certain extent, these two factors are interdependent. A faulty weapon or a poorly adjusted weapon can cause casualties in normal ammunition, but if the weapon be in proper condition when casualties are encountered, then the fault lies with the ammunition.

20.0 EQUIPMENT

Equipment listed in the Function and Casualty section of the appropriate Inspection Equipment List shall be used.

30.0 TEST PROCEDURE

30.1 Pre-firing (Preparation for test)

30.1.1 Weapons shall be of the latest design or most recent issue. Weapons shall be discarded when unusual behavior occurs which is proven attributable to the weapon.

30.1.2 Weapons employed in this test must have barrels in serviceable condition. To assure this, all barrels shall be inspected visually before use and the weapon rejected if any defect is discovered. Unusual behavior, proven to be attributable to the barrel, may also be reason for discarding the weapon.

30.1.3 The test cartridges shall be examined for obvious defects before the Function and Casualty test. If visual defects are found, the defective cartridge(s) shall be replaced and the defects shall be noted on the test sheet form.



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APPENDIX "F" (CONT'D)

30.1.4 The test weapon may be fired off-hand or in a fixed rest, as desired. Firing shall be alternated, i.e., six (6) cartridges single action, six (6) cartridges double action, six (6) cartridges single action, etc.

30.2 During Firing

30.2.1 The number of cartridges fired single and double action (6 each) shall be governed by the capacity of the cylinder of the revolvers. The weapon shall be at room temperature at the beginning of the test, and shall be cooled after firing 48 cartridges.

30.2.2 The number of revolvers to be used and the quantity of cartridges to be fired in each, shall be as specified in the applicable specification. The procedure for firing each of the weapons specified shall be the same insofar as possible. Six (6) cartridges shall be inserted into the cylinder of the weapon and fired single action; six (6) cartridges are then inserted and fired double action; this procedure is followed until the specified number of cartridges have been fired. A time interval of not more than one minute shall be allowed to reload the weapon each time.

30.2.3 Fired cases shall be visually examined by the technician for possible case casualties.

40.0 RECORDING OF DATA

40.1 Casualties shall be reported in accordance with terminology specified in 60.0 and the applicable specification.

40.2 Misfires shall be recorded and the cause described (see 50.2).

40.3 The Function and Casualty Test requires careful attention and alertness, and any unusual occurrences in weapon function or appearance of fired cases shall be noted.

40.4 Failures of gun parts shall be shown on the ammunition report.

50.0 OPERATIONAL NOTES

50.1 In the event any stoppage occurs during firing of the test, a detailed check shall be made to determine whether the ammunition or the equipment is at fault. If the stoppage was caused by a misfire, the weapon shall be checked thoroughly. To assist in determining whether ammunition or equipment is responsible for a stoppage, it is good practice to test the weapon in question using ammunition of known characteristics, and to test the ammunition in question by



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APPENDIX "F" (CONT'D)

firing in another weapon of the same type. If it is established that some faulty condition of the weapon is responsible for the stoppage, the test shall be disregarded, the weapon shall be corrected or replaced, and the tests with that type weapon shall be refired.

50.2 If a misfire is encountered, irrespective of the type of test wherein it occurs, the weapon is examined carefully to determine if the cause if attributable to the weapon. In any ballistic acceptance test where a misfire occurs, a second attempt to fire a primer shall not be made. It is mandatory that a period of at least five (5) minutes elapse after the misfire occurs before removing the cartridge from the weapon. Existing safety regulations shall be exercised when removing the cartridge from the chamber; the cartridge is then preserved for further examination. All handling and examinations of misfired cartridges shall be conducted with due regard for the hazards involved. The weapon in which a misfires occurs shall be checked thoroughly: it shall be disassembled and all component parts critically scrutinized. Results of such examination shall be included on the test report as a matter of information.

50.2.1 Laboratory examination of the misfired cartridges shall be made to determine the specific cause; the result of the investigation is included on the test report.

50.3 Upon completion of firing, all cartridge cases from the test ammunition shall be carefully examined for firing defects. If any defect is found, a detailed check of the equipment shall be made to determine whether the ammunition or the equipment is at fault. If it is established that a faulty weapon is responsible for the firing defect, then the test shall be disregarded, the weapon shall be corrected or replaced, and the tests with that type weapon shall be refired. If it cannot be established that the weapon or other equipment is at fault, then the firing defects shall be charged against the ammunition.

60.0 DEFINITIONS

60.1 Misfire.-Failure of a round of ammunition to fire after the initiating impulse has been applied to the primer.

a. The primer fails to fire when struck by the firing pin.

b. The propellant does not ignite when the primer fires normally.

60.2 Hangfire.-Any delay in the functioning of a cartridge after the initiating impulse has been applied to the primer.



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APPENDIX "F" (CONT'D)

60.3 Pierced Primer.-A pierced primer is one in which the primer cup is entirely perforated by the firing pin. It can be identified by a visible hole through the primer, or, if the perforation be minute, by discoloration of the firing pin indent caused by burning gas. This casualty may be caused by an imperfect firing pin or by one of improper contour; or it may be caused by the metal in the base of the primer cup being too thin, or too soft, or defective from laminations, scale, etc.

60.4 Primer leak.-Discoloration caused by gas leakage around the junction between the primer cup and the primer pocket wall.

60.5 Loose Primer.-Looseness which permits the fired primer to fall from the primer pocket after the cartridge is fired, without distortion to the primer pocket.

60.6 Blown Primer.-A blown primer is a primer which, when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and pocket are enlarged and deformed. This condition results from a soft head.

60.7 Flow-Back.-A flow-back may be described as the distortional forcing of the dome of the primer cup into the space between the firing pin and the firing pin hole in the bolt face.

60.8 Punch-Out.-A punch-out is the removal of metal in the dome of the primer.

60.9 Primer Setback.-The backward movement of a primer cup in a cartridge case which occurs when the base of the cup is not properly supported by the bolt face, and protrudes above the head of the cartridge case.

60.10 Ruptured Case.-A circumferential separation of the case wall produced by firing. Ruptures are divided into two categories, partial and complete. A partial rupture is one which extends less than 360 degrees around the case; a complete rupture is one which extends entirely around the case, separating the case into two parts. Ruptures are designated according to position, as indicated on Drawing B7643674.

60.11 Split Case.-A longitudinal separation of the metal in the case wall produced by firing. Splits shall be classified as prescribed by the cartridge specification and Drawing C7643674.

60.12 Rim Shear.-A case which exhibits complete removal of any portion of the extractor rim during normal gun functioning, usually resulting in failure to extract.



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APPENDIX "F" (CONT'D)

60.13 Failure to Extract.-The fired case is not removed from the weapon chamber by normal gun action.

60.14 Fouling.-Deposit that remains in the bore of a gun after it is fired.

60.15 Stripped Jacket.-The jacket of the bullet separates from the core after it has left the muzzle of the weapon, or separates in the bore and portions of jacket remain in the bore.



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## APPENDIX "G"

## TEMPERATURE (HIGH AND LOW) TEST PROCEDURE

10.0 PURPOSE

To determine the ballistic stability of ammunition after being stored at high or low temperature.

20.0 EQUIPMENT

20.1 Equipment listed in Temperature Test section of the appropriate Inspection Equipment List shall be used.

20.2 Temperature-controlled container or containers capable of maintaining temperatures within a tolerance of $\pm$ 2°F. (insulated boxes which have been brought to the required temperature may be used to transfer ammunition from controlled-temperatures storage to the test weapons if portable controlled-temperature containers are not available.

20.3 The temperature-controlled container(s) shall be of sufficient capacity to permit conditioning the ammunition to the desired temperature, and to allow free circulation of air around the cartridges.

30.0 TEST PROCEDURE

30.1 Pre-firing (Preparation for test)

The same procedures used in preparation for firing Chamber Pressure (Appendix "D") and Function and Casualty (Appendix "F") shall be used for this test.

30.1.1 For purposes of safety, the Chamber Pressure test shall be conducted prior to firing the Function and Casualty Test. If the chamber pressure of any individual cartridge exceeds the limit set in the applicable specification, the Function and Casualty test shall not be conducted.

30.1.2 Temperature of test ammunition shall be 60°to 80°F., prior to conditioning at high or low temperatures. The ammunition shall be placed in the temperature-controlled container(s) in such a manner that all cartridges are subjected to a uniform temperature, and held thereat for the time specified.

30.2 During Firing



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APPENDIX "G" (CONT'D)

30.2.1 The ammunition under test, stored at the temperature specified in the applicable specification is removed from the controlled temperature container, and placed in the insulated box (referenced in 20.2), which has also been conditioned to the temperature specified. The insulated box is then placed at a convenient point to the technician.

30.2.2 The procedures used for firing Chamber Pressure Test (Appendix "D") and Function and Casualty Test (Appendix "F") shall be used for this test with the following exceptions:

For Chamber pressure firing, three cartridges are removed from the controlled temperature container and placed in the insulated box. The cartridges are then removed singly from the insulated box and fired.

For Function and Casualty firing, six cartridges are removed from the controlled temperature container and placed in the insulated box. The cartridges are then loaded in the revolver and fired as specified.

These procedures shall be followed for either high or low temperature tests.

40.0 RECORDING OF DATA

40.1 Results shall be recorded directly on the test report form as follows:

40.1.1 Chamber pressure.-Individual velocities and chamber pressures, average velocity and chamber pressure, extreme variation of velocity and chamber pressure, chamber pressure correction, average chamber pressure (corrected), standard deviation of the individual chamber pressures.

40.1.2 Function and Casualty.-Number and type of case casualties and weapon malfunctions.

40.1.3 Temperature at which test cartridges were stored and fired.

40.1.4 The following test weapon data shall be recorded on the test report form:

a. Receiver number

b. Barrel number

c. Total number of cartridges fired in barrel



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## APPENDIX "H"

## WATERPROOF TEST PROCEDURE

10.0 PURPOSE

To determine the water tightness of cartridges.

20.0 EQUIPMENT

Equipment shall be the equivalent of that shown on drawing listed in the Waterproof section of the appropriate Inspection Equipment List.

30.0 TEST PROCEDURE

30.1 Preparation for Test

30.1.1 Convenient configuration for the reservoir and glass test chamber are shown on the drawing referenced in 20.0. The glass test chamber should be at least of sufficient diameter to accomodate five cartridges lying horizontally on a perforated rack or tray, and of sufficient depth to allow at least two inches of water above the cartridges for the test. The volume of the reservoir should be about five to fifteen times the volume of air remaining in the glass test chamber when the desiccator contains sufficient water for conducting the test.

30.1.2 The glass test chamber shall contain a sufficient amount of freshly boiled water (cooled to room temperature) to allow a 2 to 2½ inch head of water above the cartridges. A perforated metal tray shall be placed across the narrow part of the body. The ground glass surfaces on the lid and body and on the glass top shall be smeared with vaseline or vacuum grease.

30.2 Conducting the Test

30.2.1 With the glass test chamber closed to the vacuum reservoir and the valve open from the reservoir to the vacuum pump, the vacuum pump shall be set in operation. The pressure in the reservoir shall be reduced until the vacuum gage shows some predetermined reading greater than that specified in the specification; this predetermined reading is the vacuum in the reservoir which will, when the reservoir is connected to the glass test chamber, produce the desired vacuum in both. (The desired vacuum in the reservoir will be approximately one to three inches greater than that required by the specification and should be determined by experiment with the particular apparatus to be used, and recorded for future reference.) When the desired vacuum in the reservoir has been obtained, the valve from the reservoir to the vacuum pump shall be closed, and the pump operation may be stopped.



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APPENDIX "H" (CONT'D)

30.2.2 The ammunition to be tested (not exceeding the number specified) shall be placed horizontally on the tray in the glass test chamber and the lid placed in position. The glass tap shall be turned to allow the vacuum reservoir to evacuate the glass test chamber to the required pounds per square inch below atmospheric pressure and shall be held at that pressure for the specified time. The number of bubbles liberated from the mouth or primer, or both, of each cartridge shall be observed. At the end of the specified time, the vacuum shall be released from the glass test chamber, the lid removed and the ammunition removed.

30.2.3 The procedure prescribed in 30.2.1 and 30.2.2 shall be repeated until the required number of cartridges has been tested.

40.0 RECORDING OF DATA

Results of Waterproof test shall be reported as follows:

40.1 No leak

40.2 Slow leak.-A series of two or more air bubbles appearing at the primer or mouth of the case, or both, but which are liberated at such a rate that only one bubble from either the primer or the mouth of the cartridge is in transit to the surface at any one time.

40.3 Fast leak.-A series of air bubbles appearing at the primer or mouth of the case, or both, which are liberated at such a rate that more than one bubble from the primer or the mouth of the cartridge case is in transit to the surface at any one time.

40.4 Location of leak; identify area of leak with machinist layout ink.



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## APPENDIX "I"

## CARTRIDGE ELONGATION TEST PROCEDURE

10.0 PURPOSE

When firing ammunition in a revolver, the recoil of the weapon tends to cause the bullets in the unfired cartridges to move forward causing the cylinder to jam, preventing it from revolving. The Cartridge Elongation test is fired to determine if the bullets contained in the test ammunition are held in the cartridge cases in such a manner to prevent their forward movement, when confirmed in the cylinder of the revolver while other cartriges are being fired.

20.0 EQUIPMENT

20.1 Equipment listed in the Cartridge Elongation section of the appropriate Inspection Equipment List.

20.2 Firing range with suitable bullet-stop.

30.0 TEST PROCEDURE

30.1 Pre-firing (Preparation for test)

30.1.1 The number of cartridges as prescribed in the applicable specification are placed in a recessed holding block in such a manner that their identity can be maintained throughout the test.

30.1.2 The overall length of each cartridge is carefully measured and the measurements recorded on the report form. The cartridges are then returned to the recessed holding block to their original position.

30.2 During Firing

30.2.1 One identified and measured cartridge from the test sample shall be placed in the cylinder of the test weapon. Five unmeasured cartridges shall be placed in the cylinder of the weapon. The cartridges are positioned in the cylinder of the weapon in such a manner that the five unmeasured cartridges will be fired; the measured test cartrige shall not be fired.

30.2.2 The five unmeasured cartridges are then fired from a hand-held revolver and the fired cases removed from the cylinder.

30.2.3 The identified cartridge shall then be removed from the cylinder and placedin its original position in the recessed holding block.



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APPENDIX "I" (CONT'D)

30.2.4 The procedure prescribed in 30.2.1 through 30.2.3 shall be repeated for each cartridge in the test sample.

30.2.5 After all cartridges in the test sample have been tested, the overall length of each of the test cartridges is carefully measured again and the measurements placed on the report form in such a manner that the measurements can be associated with the initial measurements.

40.0 RECORDING OF DATA

Test sheets shall show the following:

a. Measurements of the overall length of the individual test cartridges, both before and after the test.

b. Elongation, if any has taken place.

40.1 Test weapon data

a. Weapon number

b. Total number of cartridges fired in weapon.



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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-46409C(MU) Cartridge, Caliber .38, Special, Ball, M41A1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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